

Annapurna Modern Data Stack - Question 2

Task-wise Outputs, Explanations, Reasons and Screenshots

This separate report documents Question 2, parts (a) to (e), using the available project outputs and screenshots. The section names should be matched with the official assignment wording if it differs.

Question 2(a): Data Validation and Verification

Task: Verify the processed dataset and confirm that the generated analytical files are available.

Output:

Raw rows processed: 1,137,585
Duplicates removed: 16,661
Cleaned file: sales_cleaned.parquet
Summary files: daily_sales.csv and monthly_sales.csv

Explanation: The processed dataset and analytical files were generated successfully. The duplicate-removal figure records the cleaning performed.

Reason for the output: Validation ensures that repeated or invalid records do not affect analysis and confirms that the required files are ready.

Directory: C:\Users\ub02-glab-012\Downloads\Q1_Annapurna\output

Mode	LastWriteTime		Length	Name
d----	18-09-2026	15:27		partitioned
d----	18-09-2026	15:27		star_schema
-a----	18-09-2026	15:27	274432	annapurna.duckdb
-a----	18-09-2026	15:32	11708	daily_sales.csv
-a----	18-09-2026	15:32	430	monthly_sales.csv
-a----	18-09-2026	15:27	209791	price_revisions.csv
-a----	18-09-2026	15:27	102611	price_revisions.parquet
-a----	18-09-2026	15:27	106629	products.csv
-a----	18-09-2026	15:27	32616	products.parquet
-a----	18-09-2026	15:27	472	product_categories.csv
-a----	18-09-2026	15:27	3196	product_categories.parquet
-a----	18-09-2026	15:32	21509063	sales_cleaned.parquet
-a----	18-09-2026	15:27	1299	stores.csv
-a----	18-09-2026	15:27	6044	stores.parquet

Figure: Generated project outputs

Question 2(b): Database and Table Verification

Task: Verify the PostgreSQL database and confirm that the required tables contain data.

Output:

stores: 12 rows
product_categories: 14 rows
products: 1,224 rows
price_revisions: 4,320 rows

Explanation: The table counts confirm that PostgreSQL contains the required data structures and records.

Reason for the output: Database verification helps detect incomplete ingestion or missing data before analytical queries are executed.

Question 2(c): Object Storage Verification

Task: Verify that project files were stored in MinIO.

Output:

Bucket: annapurna
Objects stored: 172
Approximate storage: 141.7 MiB

Explanation: The MinIO bucket contains stored project objects, including raw files, processed data and reports.

Reason for the output: Object storage is used for the data-lake layer because it can store different file formats independently of the relational database.

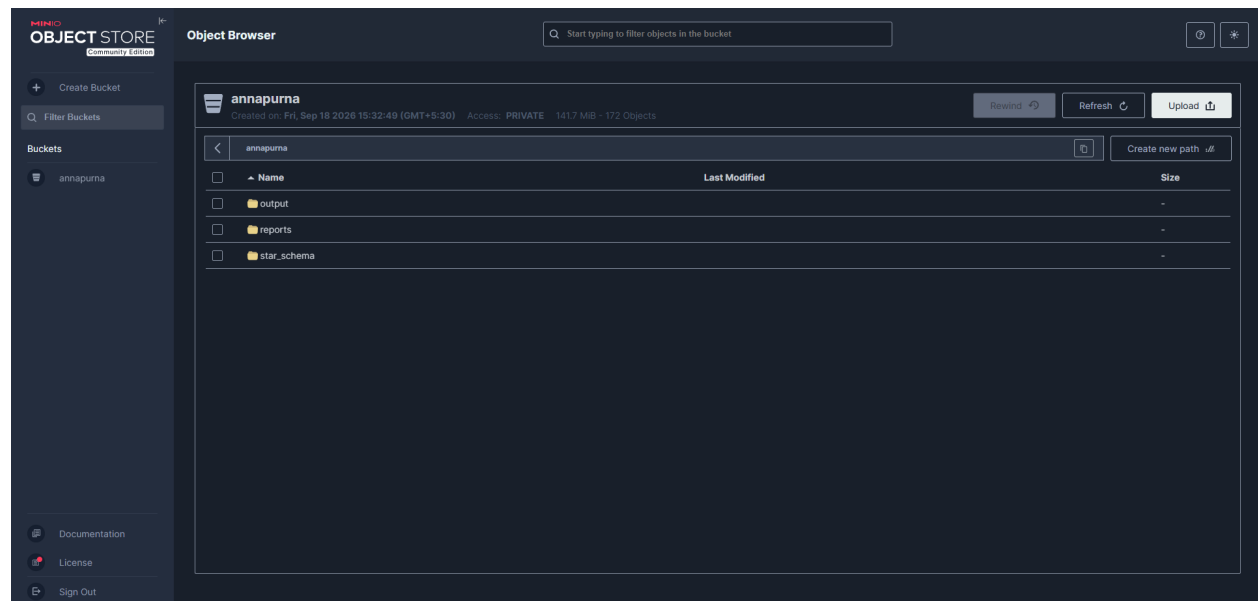


Figure: MinIO bucket and stored project objects

Question 2(d): Analytical Query and Curated Output Generation

Task: Use DuckDB to process the data and generate analytical outputs.

Output :
DuckDB database: annapurna.duckdb
Outputs: daily_sales.csv, monthly_sales.csv, sales_cleaned.parquet and partitioned analytical data.

Explanation: The output shows that analytical processing generated a DuckDB database and reusable summary files.

Reason for the output: DuckDB supports local analytical processing and efficient queries over CSV and Parquet data.

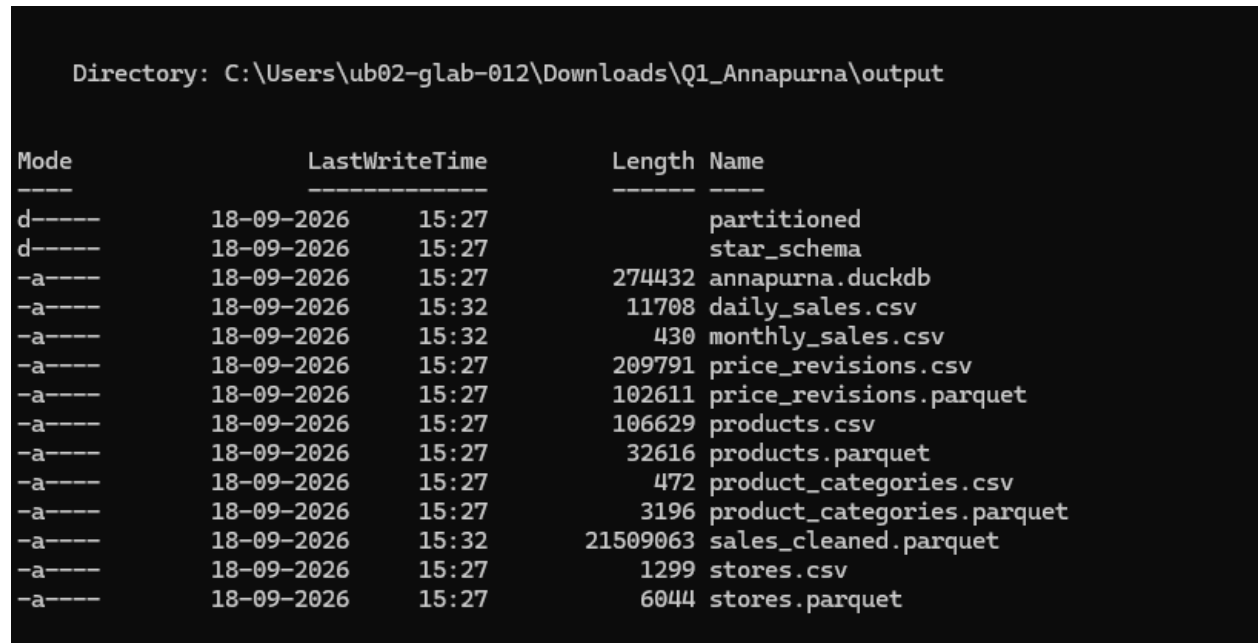


Figure: Generated project outputs

Question 2(e): Final Data Products and Reconciliation

Task: Prepare the final analytical files and generate the finance reconciliation output.

Output:

Files include annapurna.duckdb, daily_sales.csv, monthly_sales.csv, finance_reconciliation.csv, sales_cleaned.parquet, reference CSV files, partitioned/ and star_schema/.

Explanation: The final data products contain the database, cleaned data, summaries, reference tables, partitioned data and star-schema outputs.

Reason for the output: Reconciliation provides a check on calculated values. Organizing the outputs together makes the project easier to evaluate and push to GitHub.

```
annapurna=# \q
What's next:
  Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug annapurna\_postgre
  Learn more at https://docs.docker.com/go/debug-cli/
PS C:\Users\ub02-glab-012\Downloads\Q1_Annapurna> Get-Content .\reports\finance_reconciliation.csv
month,calculated_revenue
2024-01,41629544.81
2024-02,37539932.03
2024-03,45168740.99
2024-04,40986594.69
2024-05,44865035.18
2024-06,42052907.32
2024-07,43292272.79
2024-08,48690619.05
2024-09,48161223.24
2024-10,60718437.18
2024-11,55888183.67
2024-12,54640017.72
PS C:\Users\ub02-glab-012\Downloads\Q1_Annapurna> |
```

Figure: Finance reconciliation output

Conclusion

Question 2 is documented through data validation, database verification, object storage, analytical processing and final data products. The included screenshots support the available outputs.